

Forward KL Can Be Better: Cross-Regularization Loss for Normalizing Flow Boltzmann Sampling

Qi Feng¹, Rongjie Lai², Di Qi², and Xuda Ye²

¹Department of Mathematics, Florida State University, Tallahassee, FL 32306

²Department of Mathematics, Purdue University, West Lafayette, IN 47907

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Abstract

Training normalizing flow Boltzmann generators with the forward KL is fragile in three complementary ways: it is silent about modes the flow misses (the effective sample size can stay near one while a large fraction of the target mass is absent), it cannot see off-support leakage in regions the target excludes, and it oscillates transversally on thin low-dimensional manifold supports. We address all three by augmenting the forward KL with a family of cross-regularization functionals X_ω that penalize the pairwise variation of $\log(\mu/\nu)$ under a non-negative weight ω . Three concrete instances are studied, X_μ for in-mode shape control, $X_{\hat{\mu}}$ for stable mode discovery against a wide-coverage measure $\hat{\mu}$, and $X_{\bar{\nu}}$ for off-support leakage suppression, together with a balanced mixture that combines $\hat{\mu}$ and $\bar{\nu}$ inside a single pairwise integral. A Fisher–Rao analysis shows that the forward KL augmented with X_μ decays exponentially with no structural assumption on μ , and that the asymptotic accuracy floor under biased AIS samples is strictly improved. The framework is conservative on low-dimensional manifold targets with roughly uniform along-manifold density. Heavy-tailed manifold targets remain hard: the quench-and-temper mode-finder that supplies $\hat{\mu}$ tends to miss the thin outer tails, and the regularizer inherits the blind spot.

1 Introduction

Drawing samples from a target distribution $\mu \propto \exp(-U)$ on $\mathbb{R}^d$ known only up to its normalization constant arises across molecular simulation, Bayesian inference, and statistical physics, and is widely tackled with Boltzmann generators built on normalizing flows (Rezende and Mohamed, 2015; Papamakarios et al., 2017; Kingma and Dhariwal, 2018; Noé et al., 2019; Wu et al., 2020). A flow is an invertible parametric map $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ trained so that the pushforward $\nu := G_{\#}^{-1}\mu_0$ of an easily samplable reference μ_0 (typically a Gaussian) approximates μ . The trained flow returns independent samples from ν along with an exact, tractable density, and importance reweighting against μ then converts those into Monte Carlo estimates with controllable finite-sample error.

Two training objectives dominate the literature. The original Boltzmann generator (Noé et al., 2019) and its variational-inference predecessor (Rezende and Mohamed, 2015) minimize the reverse KL

$$\text{KL}(\nu\|\mu) = \int_{\mathbb{R}^d} \nu(x) \log \frac{\nu(x)}{\mu(x)} dx, \quad (1)$$

which needs only samples from ν and is therefore cheap to estimate. It is, however, mode-seeking: on multimodal μ the flow tends to collapse onto a subset of the modes from which it cannot recover (Noé et al., 2019; Wu et al., 2020). The opposing family minimizes the forward KL (Gabrié et al., 2022; Wirsberger et al., 2020),

$$\text{KL}(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx, \quad (2)$$

which is mode-covering but needs samples from μ . Such samples are produced by annealed importance sampling (AIS) (Neal, 2001), which transports ν toward μ through a ladder of intermediate distributions while accumulating importance weights. The flow-AIS bootstrap (FAB) of Midgley et al. (2022) couples AIS to flow training and replaces the target by the mode-sensitive surrogate μ^2/ν to encourage exploration.

A separate line of work drops the explicit-density requirement and trains a diffusion- or SDE-based sampler directly: the path-integral sampler (Zhang and Chen, 2022), denoising diffusion samplers (Vargas et al., 2023), and the more recent iterated denoising / energy-matching family (Akhound-Sadegh et al., 2024; Woo and Ahn, 2024) learn a stochastic process whose terminal distribution matches μ . These samplers are expressive and scale well, but they do not deliver an exact tractable density of their output, so importance reweighting against μ is no longer available. We commit to flow-based methods for that reason: the invertibility of G gives an explicit ν and therefore an exact, finite-sample importance-sampling pipeline, in which the effective sample size (ESS) (Liu, 2001; Doucet et al., 2001; Martino et al., 2017) computed from the flow’s own weights is a standard and convenient diagnostic of training quality, with known blind spots that motivate the failure-mode analysis below.

Forward KL training has three complementary failure modes that we use to organize the rest of the paper. The first is silent missed modes, the fake ESS pitfall: ESS is gold-standard only when μ and ν are evaluated on the same support, so if ν misses modes of μ , the AIS chain initialized at ν rarely escapes the basin ν already covers (Noé et al., 2019; Midgley et al., 2022) and the resulting ESS can be close to 1 while a large fraction of the mass of μ is missing. The second is off-support leakage. Where μ vanishes, the forward KL integrand $\mu \log(\mu/\nu)$ also vanishes, so the gradient is blind to mass that ν places outside the support of μ . The third is transversal oscillation on low-dimensional manifolds. When μ concentrates on a thin manifold, the forward KL controls only the in-manifold marginal of $\log(\mu/\nu)$, and the off-manifold profile of ν is left free to oscillate. All three blind spots persist no matter how accurately AIS samples the modes ν already reaches.

Our remedy is to augment the forward KL with a family of cross-regularization functionals that penalize variation of the log-ratio $\log(\mu/\nu)$ directly. Let

$$X_\omega(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \omega(x, y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (3)$$

where ω is a non-negative measure on $\mathbb{R}^{2d}$. The motivation is direct: matching ν to μ is the same as flattening $\log(\mu/\nu)$, and the pairwise difference cancels the unknown normalization of μ . Suitably localized choices of ω provide the three safeguards corresponding to the three failure modes above: stable mode discovery (against silent missed modes), off-support leakage suppression (against gradient blindness off the support of μ), and shape accuracy on low-dimensional manifolds (against transversal oscillation). As a loss ingredient X_ω has three properties that we use throughout. First, it gives pointwise control: the forward KL constrains $\nu \approx \mu$ only in a global Jensen-averaged sense, whereas X_ω penalizes log-ratio variation wherever ω places mass and localizes the constraint to the support of ω . Second, it is robust to the sampling of ω : X_ω vanishes whenever $\log(\mu/\nu)$ is constant on the support of ω , so its minimizer is insensitive to the precise shape of ω , and even hand-injected mode locations can be added to ω without corrupting it. Third, it is compatible with the forward KL: for $\omega = \mu \otimes \mu$ the integrand of X_ω coincides with that of the forward KL up to a sign, so the same autograd graph and the same μ -samples are reused at negligible extra cost, the scales match, and no extra hyperparameter is needed.

Each choice of ω localizes the constraint to a region of $\mathbb{R}^d$; we use three choices with complementary roles. With $\omega = \mu \otimes \mu$,

$$X_\mu(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x)\mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (4)$$

which is essentially free by the forward KL compatibility above. The forward KL controls the μ -mean of $\log(\mu/\nu)$ while X_μ controls its μ -spread, tightening the in-mode density match once ν has reached the modes of μ and reducing the residual bias inherited from imperfect AIS samples.

With $\omega = \hat{\mu} \otimes \hat{\mu}$,

$$X_{\hat{\mu}}(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \hat{\mu}(x)\hat{\mu}(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (5)$$

where $\hat{\mu}$ is a coarse, deliberately over-spread approximation of μ . By sample robustness the minimizer of $X_{\hat{\mu}}$ is insensitive to the precise shape of $\hat{\mu}$, so the sole purpose of $\hat{\mu}$ is to place mass on modes and tails that the current ν has not yet covered; the gradient then pulls the flow toward those regions. We build $\hat{\mu}$ once, before training, with a quench and temper (QT) algorithm that diffuses an initial cloud through Brownian noise to bridge basins, runs a deterministic mode-finder to collapse it onto mode centers, and rejuvenates with a short Langevin chain. QT errs on the side of broad coverage and works well on targets with disconnected modes and on thin low-dimensional manifolds whose density along the manifold is roughly uniform. It has a known weakness on heavy-tailed targets: the thin, low-mass outer tail is easily missed because the diffusion step rarely scatters particles close enough, and the deterministic mode-finder pulls any near miss back into the wider basins of the body.

With $\omega = \bar{\nu} \otimes \bar{\nu}$,

$$X_{\bar{\nu}}(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \bar{\nu}(x)\bar{\nu}(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (6)$$

where $\bar{\nu}$ denotes a stop-gradient copy of the current pushforward ν . Sampling from $\bar{\nu}$ is free, since the training loop already pushes a batch from μ_0 through G^{-1} at every step; the detached gradient keeps the loss numerically stable on sharp targets where a true reverse KL gradient explodes. $X_{\bar{\nu}}$ plays the reverse KL role without its pathologies: it penalizes mass that ν places outside the support of μ , suppressing the mode-overfitting leakage that X_{μ} and $X_{\hat{\mu}}$ alone cannot see.

The three weights are complementary: X_{μ} refines what AIS already reaches, $X_{\hat{\mu}}$ pulls ν toward modes it has missed, and $X_{\bar{\nu}}$ pushes ν away from regions outside the support of μ . Our preferred training loss combines all three with the forward KL,

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_{\mu}(\mu\|\nu) + X_{(\alpha\hat{\mu}+\beta\bar{\nu}) \otimes (\alpha\hat{\mu}+\beta\bar{\nu})}(\mu\|\nu), \quad (7)$$

where $\lambda, \alpha, \beta \geq 0$ are tunable weights and the third term is X_{ω} with mixture weight $\omega = \alpha\hat{\mu} + \beta\bar{\nu}$. This mixture form is strictly stronger than the sum $\alpha^2 X_{\hat{\mu}} + \beta^2 X_{\bar{\nu}}$: the pairwise integral also generates cross contributions $|\log(\mu/\nu)(x) - \log(\mu/\nu)(y)|$ with $x \sim \hat{\mu}$ and $y \sim \bar{\nu}$, which anchor the in-support and out-of-support log-ratios to each other. We confirm this empirically: across our 2D benchmarks the mixture variant attains the best full-coverage ESS while always reaching complete mode coverage.

Equation (7) is written in the abstract density ν , but the trainable object is the flow map G . By the change of variables $\nu(y) = \mu_0(G(y))|\det J_G(y)|$, and writing $U_0 = -\log \mu_0$ for the source potential, the log-ratio along the flow is the single field

$$\log \frac{\mu(y)}{\nu(y)} = U_0(G(y)) - U(y) - \log |\det J_G(y)| =: z(y),$$

up to an additive constant. Substituting into (7) expresses the preferred loss directly as an objective in G ,

$$\mathcal{L}[G] = \underbrace{\mathbb{E}_{y \sim \mu} [z(y)]}_{\text{KL}(\mu\|\nu)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu} [|z(y) - z(y')|]}_{X_{\mu}(\mu\|\nu)} + \underbrace{\mathbb{E}_{y, y' \sim \alpha\hat{\mu} + \beta\bar{\nu}} [|z(y) - z(y')|]}_{X_{\alpha\hat{\mu} + \beta\bar{\nu}}(\mu\|\nu)}, \quad (8)$$

which is the loss we minimize by stochastic gradient descent and the central object of this paper. Every term is read off a single forward pass of G on a batch – the forward KL is the batch mean of z , and each X term the mean absolute pairwise difference of z over the batch – so the regularizers cost essentially nothing beyond a vanilla forward KL step. Section 2 analyzes the Fisher–Rao gradient flow of this loss, and Section 3 details the annealed importance sampling and quench-and-temper sampling that supply its three expectations.

We pair ESS with the coverage metric of [Naeem et al. \(2020\)](#) for evaluation, which counts the fraction of QT-discovered $\hat{\mu}$ samples whose k -nearest-neighbor ball is reached by the trained ν . Coverage exposes the modes that a fake ESS hides, so the two metrics together certify both density calibration and mode coverage.

This paper makes three contributions. We introduce the cross-regularization framework X_{ω} , study three concrete instances X_{μ} , $X_{\hat{\mu}}$, $X_{\bar{\nu}}$, and identify the mixture variant $X_{(\alpha\hat{\mu}+\beta\bar{\nu}) \otimes (\alpha\hat{\mu}+\beta\bar{\nu})}$ that combines mode discovery and leakage suppression inside a single pairwise integral. We carry out a Fisher–Rao convergence analysis showing that the forward KL augmented with X_{μ} decays exponentially with no structural assumption on μ , and that the asymptotic accuracy floor under biased AIS samples is strictly improved by the X_{μ} term. We report an empirical study on six 2D benchmarks (Threewell, Himmelblau, Annulus, Python, Chessboard,

Sparse), where, among the four losses we compare, the mixture variant is the only one that both matches or beats the best full-coverage ESS and reaches full mode coverage on every target, together with a high-dimensional product multi-well family up to $d = 128$, where the gap between the X-augmented losses and the bare forward KL widens with the dimension.

2 Fisher–Rao gradient flow with X functionals

Before turning to the training algorithm, we study the gradient-flow properties of the loss on the space of densities; the convergence rate of this continuous-time idealization is a clean proxy for how trainable the corresponding discrete loss is in practice. We view ν as a generic probability density evolving along the gradient flow of the loss under an appropriate Riemannian metric on the space of densities (Ambrosio et al., 2008; Villani, 2009) and ask how fast ν_t converges to the target μ in $\text{KL}(\mu\|\nu_t)$.

In this section we restrict to the loss with only the forward KL and the raw-target regularizer X_μ ,

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x) \mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (9)$$

where $\lambda > 0$ is a fixed constant. Throughout this section we abbreviate the importance weight by

$$w(x) := \frac{\mu(x)}{\nu(x)},$$

so that the log-ratio is $\log w(x) = \log \mu(x) - \log \nu(x)$.

The choice of Riemannian metric matters. In the Wasserstein-2 geometry, induced by optimal transport (Jordan et al., 1998; Otto, 2001), the forward KL fails to be displacement convex along Wasserstein-2 geodesics, and the standard Otto–Villani / Bakry–Émery machinery (Otto and Villani, 2000; Bakry et al., 2014) that yields exponential convergence for the reverse KL does not apply. We therefore work in the Fisher–Rao geometry (Amari, 2016; Bauer et al., 2016), in which the forward KL admits an unconditional exponential decay rate and the regularizer X_μ contributes a qualitatively new, nonlocal birth–death modulation (Lu et al., 2019): the local update at x is driven by how the importance weight at x ranks against the population of weights under μ .

Denoting the ν -mean by $\langle \cdot \rangle_\nu := \int (\cdot) \nu dx$, the Fisher–Rao gradient of a functional $\mathcal{F}[\nu]$ and the associated dissipation identity along the FR gradient flow $\partial_t \nu_t = -\text{grad}^{\text{FR}} \mathcal{F}(\nu_t)$ are

$$\text{grad}^{\text{FR}} \mathcal{F}(\nu) = \nu \left(\frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_\nu \right), \quad \frac{d}{dt} \mathcal{F}[\nu_t] = - \int_{\mathbb{R}^d} \left(\frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_{\nu_t} \right)^2 \nu_t dx. \quad (10)$$

The mean-subtraction enforces $\int \text{grad}^{\text{FR}} \mathcal{F} = 0$, so the FR flow conserves probability mass automatically.

2.1 Derivation of X_μ -augmented Fisher–Rao gradient flow

The first variation of the forward KL with respect to ν admits the simple form

$$\frac{\delta \text{KL}(\mu\|\nu)}{\delta \nu}(x) = -\frac{\mu(x)}{\nu(x)} = -w(x), \quad (11)$$

so by (10) the FR gradient of the forward KL is

$$\text{grad}^{\text{FR}} [\text{KL}(\mu\|\nu)](x) = \nu(x) - \mu(x). \quad (12)$$

For the X_μ functional, the inner log-ratio difference has first variation

$$\frac{\delta (\log w(x) - \log w(y))}{\delta \nu(x)} = \frac{1}{\nu(y)} \delta(x - y) - \frac{1}{\nu(x)}.$$

By the symmetry of X_μ in x and y ,

$$\begin{aligned}\frac{\delta X_\mu(\mu\|\nu)}{\delta\nu}(x) &= 2 \int_{\mathbb{R}^d} \mu(x)\mu(y) \operatorname{sgn}(w(x) - w(y)) \frac{\delta(\log w(x) - \log w(y))}{\delta\nu(x)} dy \\ &= 2 \int_{\mathbb{R}^d} \mu(x)\mu(y) \operatorname{sgn}(w(x) - w(y)) \left(\frac{1}{\nu(y)} \delta(x - y) - \frac{1}{\nu(x)} \right) dy \\ &= -2 \int_{\mathbb{R}^d} w(x) \operatorname{sgn}(w(x) - w(y)) dy,\end{aligned}$$

where the contribution of $\delta(x - y)$ vanishes because $\operatorname{sgn}(w(x) - w(y)) = 0$ at $y = x$. Hence

$$\frac{\delta X_\mu(\mu\|\nu)}{\delta\nu}(x) = -2w(x)S(x), \quad S(x) := \mathbb{E}_{y \sim \mu} [\operatorname{sgn}(w(x) - w(y))]. \quad (13)$$

We call S the *rank field* of the importance weight under μ : $S(x) > 0$ where $w(x)$ exceeds the μ -median of w , i.e. where ν undercovers a mode of μ ; $S(x) < 0$ where ν overcovers; and $S \equiv 0$ at $\nu = \mu$. By antisymmetry,

$$\mathbb{E}_\mu[S] = \mathbb{E}_{x, y \sim \mu} [\operatorname{sgn}(w(x) - w(y))] = 0,$$

and the FR gradient of X_μ with respect to ν is

$$\operatorname{grad}^{\text{FR}}[X_\mu(\mu\|\nu)](x) = -2\mu(x)S(x). \quad (14)$$

Since the FR gradient is linear in the functional, the FR gradient of $\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu)$ is the direct sum of (12) and (14):

$$\operatorname{grad}^{\text{FR}}[\mathcal{L}](x) = \mu(x) - \nu(x) + 2\lambda\mu(x)S(x).$$

The augmented Fisher–Rao gradient flow $\partial_t \nu_t = -\operatorname{grad}^{\text{FR}}[\mathcal{L}]$ therefore reads

$$\partial_t \nu_t(x) = \mu(x) - \nu_t(x) + 2\lambda\mu(x)S_t(x), \quad \text{where } S_t(x) := \mathbb{E}_{y \sim \mu} \left[\operatorname{sgn} \left(\frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right) \right]. \quad (15)$$

Equation (15) is a *nonlocal* PDE: its right-hand side at x depends on $S_t(x)$, which in turn requires the entire profile of the importance weight μ/ν_t over the support of μ . Mass is conserved along the flow because

$$\frac{d}{dt} \int_{\mathbb{R}^d} \nu_t(x) dx = 2\lambda \int_{\mathbb{R}^d} \mu(x) S_t(x) dx = 0,$$

and the flow is stationary at $\nu_t = \mu$, where $\mu(x)/\nu_t(x) \equiv 1$ and hence $S_t \equiv 0$. When $\lambda = 0$, (15) reduces to the pure Fisher–Rao gradient flow of the forward KL.

2.2 Convergence analysis of the X_μ -augmented flow

To analyze the convergence rate of the augmented flow (15), we take the forward KL $\text{KL}(\mu\|\nu_t)$ as the error metric and differentiate it along the flow:

$$\frac{d}{dt} \text{KL}(\mu\|\nu_t) = - \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} (1 + 2\lambda S_t) dx + \int_{\mathbb{R}^d} \mu dx = -\chi^2(\mu\|\nu_t) - 2\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx,$$

where the chi-squared divergence is

$$\chi^2(\mu\|\nu) := \int_{\mathbb{R}^d} (w - 1)^2 \nu dx = \int_{\mathbb{R}^d} w^2 \nu dx - 1.$$

The sign of the second integral is settled by the following rank-correlation identity.

Lemma 1 For any probability density ν on $\mathbb{R}^d$,

$$\int_{\mathbb{R}^d} \frac{\mu(x)^2}{\nu(x)} S(x) dx = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [|w(x) - w(y)|] \geq 0,$$

with equality if and only if w is μ -almost-everywhere constant, i.e. $\nu = \mu$.

Proof. By the definition of S ,

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \mathbb{E}_\mu [w(x) S(x)] = \mathbb{E}_{x,y \sim \mu} [w(x) \operatorname{sgn}(w(x) - w(y))].$$

Symmetrizing in $x \leftrightarrow y$ and averaging the two equal copies gives

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [(w(x) - w(y)) \operatorname{sgn}(w(x) - w(y))] = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [|w(x) - w(y)|].$$

□

Applying Lemma 1 with $\nu = \nu_t$ and $S = S_t$, the dissipation rewrites as

$$\frac{d}{dt} \operatorname{KL}(\mu \| \nu_t) = -\chi^2(\mu \| \nu_t) - \lambda \mathbb{E}_{x,y \sim \mu} \left[\left| \frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right| \right]. \quad (16)$$

The elementary inequality $w \log w \leq w^2 - w$ applied to $w = \mu/\nu_t$ gives $\chi^2(\mu \| \nu_t) \geq \operatorname{KL}(\mu \| \nu_t)$, hence $\frac{d}{dt} \operatorname{KL}(\mu \| \nu_t) \leq -\operatorname{KL}(\mu \| \nu_t)$, and Grönwall's lemma yields the exponential decay

$$\operatorname{KL}(\mu \| \nu_t) \leq e^{-t} \operatorname{KL}(\mu \| \nu_0), \quad t \geq 0, \quad (17)$$

with no structural assumption on μ : neither log-concavity, nor a spectral gap, nor any growth condition. This contrasts sharply with the Wasserstein-2 geometry, where the forward KL is not displacement convex and an analogous exponential rate is unavailable without further assumptions.

The uniform convergence rate (17) should *not* be read as a blanket statement that minimizing the forward KL is intrinsically faster or more efficient than minimizing the reverse KL. The FR flow is a continuous-time idealization on the space of densities and is agnostic to the sampling cost. In practice, the difficulty of training has been partially transported to the AIS step: evaluating the forward KL and its gradient requires accurate samples of μ , and producing such samples in high dimensions is itself a hard problem. This is the very reason that AIS is needed, and that the X_μ regularizer must compensate for modes that AIS alone fails to reach. The exponential rate (17) quantifies how fast the flow *can* converge given accurate target samples; it does not eliminate the burden of producing them.

A naive comparison of dissipation rates between KL-only and $\operatorname{KL} + \lambda X_\mu$ would also be misleading, since scaling the loss by a constant is mere time reparametrization with no qualitative gain. The substantive contribution of X_μ is structural rather than rate-based: while the pure-KL FR gradient is local in x , the rank field S_t is global, so the augmented flow (15) couples the update at x to the entire distribution of importance weights under μ . This nonlocal coupling amplifies the birth rate on undercovered regions of μ and suppresses it on overcovered ones, a global comparison that pure forward KL (which knows only whether $\mu(x)/\nu_t(x)$ exceeds or falls below 1 pointwise) cannot produce.

Remark (Monotone decay of X_μ) The regularizer X_μ also decreases monotonically along the augmented flow: a direct computation along (15) gives

$$\frac{d}{dt} X_\mu(\mu \| \nu_t) = -2 \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx - 4\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t^2 dx \leq 0,$$

with strict inequality whenever $\nu_t \neq \mu$. Thus X_μ is itself driven to zero in the long-time limit.

2.3 Asymptotic accuracy of biased forward KL training

The convergence analysis of Section 2.2 assumes that the outer expectation in $\text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu)$ is taken under the exact target μ . In practice, the algorithm only has access to an approximation $\tilde{\mu}$ of μ supplied by AIS, and the realized loss reads

$$\tilde{\mathcal{L}}[\nu] = \int_{\mathbb{R}^d} \tilde{\mu}(x) \log w(x) dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) |\log w(x) - \log w(y)| dx dy, \quad (18)$$

where the pointwise ratio $w(x) = \mu(x)/\nu(x)$ is still evaluated against the exact target, since it is computed per sample from the target potential U and the flow’s own density. Repeating the derivation of Section 2.1 with μ replaced by $\tilde{\mu}$ in the outer integrands gives the Fisher–Rao gradient

$$\text{grad}^{\text{FR}}[\tilde{\mathcal{L}}](x) = \nu(x) - \tilde{\mu}(x)(1 + 2\lambda\tilde{S}(x)), \quad \tilde{S}(x) := \mathbb{E}_{y \sim \tilde{\mu}}[\text{sgn}(w(x) - w(y))].$$

A stationary point ν^* of (18) therefore satisfies the self-consistent equation

$$\nu^*(x) = \tilde{\mu}(x) \left[1 + 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(y) \text{sgn}\left(\frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)}\right) dy \right]. \quad (19)$$

At $\lambda = 0$ the equation collapses to $\nu^* = \tilde{\mu}$: pure forward KL training inherits the bias of the AIS distribution verbatim, and the residual accuracy is exactly $\text{KL}(\mu\|\tilde{\mu})$. For $\lambda > 0$, however, the rank-based correction $1 + 2\lambda\tilde{S}^*$ inflates ν^* above $\tilde{\mu}$ where the importance weight μ/ν^* is high (i.e. where ν^* still undercovers μ) and deflates it where the weight is low, providing a global feedback that pulls ν^* from $\tilde{\mu}$ back toward μ . This is quantified in the following theorem.

Theorem 1 *Assume μ is absolutely continuous with respect to $\tilde{\mu}$ and $\lambda \in [0, 1/2]$, so that the bracket in (19) is non-negative and ν^* is a probability density. Any stationary point ν^* of the biased loss (18) satisfies*

$$\text{KL}(\mu\|\nu^*) \leq \text{KL}(\mu\|\tilde{\mu}) - \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy. \quad (20)$$

Proof. Substituting the stationary identity $\nu^* = \tilde{\mu}(1 + 2\lambda\tilde{S}^*)$ into the difference of forward KLs and using $\mu = \nu^* \cdot (\mu/\nu^*) = \tilde{\mu}(1 + 2\lambda\tilde{S}^*)(\mu/\nu^*)$,

$$\text{KL}(\mu\|\tilde{\mu}) - \text{KL}(\mu\|\nu^*) = \int_{\mathbb{R}^d} \mu(x) \log(1 + 2\lambda\tilde{S}^*(x)) dx = \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} u(x) \log u(x) dx,$$

where $u(x) := 1 + 2\lambda\tilde{S}^*(x) \geq 1 - 2\lambda \geq 0$ since $\lambda \leq 1/2$. The elementary inequality $u \log u \geq u - 1$ for $u \geq 0$ then yields

$$\text{KL}(\mu\|\tilde{\mu}) - \text{KL}(\mu\|\nu^*) \geq 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx.$$

Applying the rank-correlation argument of Lemma 1 verbatim with μ replaced by $\tilde{\mu}$ in the outer measure,

$$\int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx = \frac{1}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy,$$

which gives (20). $\square$

The right-hand side of (20) contrasts the two roles of X_μ training. At $\lambda = 0$ the bound is tight at $\nu^* = \tilde{\mu}$ and the residual KL is fixed at $\text{KL}(\mu\|\tilde{\mu})$; the AIS bias is a hard floor. Switching on the regularizer with $\lambda \leq 1/2$ strictly lowers this floor by an amount controlled by the discrepancy of the importance weights μ/ν^* across the support of $\tilde{\mu}$: the larger this discrepancy, the more the rank-based feedback pulls ν^* toward μ . The X_μ regularizer is therefore not merely a numerical stabilizer for the forward KL flow: it directly improves the asymptotic accuracy of the trained density when the target samples are biased, which is the practically relevant regime.

The threshold $\lambda \leq 1/2$ is a worst-case condition enforcing positivity of ν^* uniformly in x ; in practice the accuracy improvement persists well beyond it. The practical takeaway is that even without the wide-coverage regularizer $X_{\tilde{\mu}}$, adding X_μ alone to the forward KL loss already reduces the bias floor inherited from inaccurate AIS samples.

3 Flow training with X functionals

This section turns the abstract loss of the introduction into a concrete training algorithm for the flow map G . Let U_0 and U be the source and target potentials on $\mathbb{R}^d$, so that $\mu_0 \propto \exp(-U_0)$ is the easily samplable reference and $\mu \propto \exp(-U)$ is the target. We parameterize $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ so that $G_{\#}\mu \approx \mu_0$ – mapping target to source, so that no inversion is required during training – and write $\nu := G_{\#}^{-1}\mu_0$ for the induced approximation to μ . We first assemble the training loss and read off the two sampling ingredients it requires (Section 3.1), then supply them in turn: annealed importance sampling of μ (Section 3.2), the quench and temper construction of the wide-coverage measure $\hat{\mu}$ (Section 3.3), and finally the ESS and coverage diagnostics used to evaluate the trained flow (Section 3.4).

3.1 The cross-regularized training loss

By the change-of-variables formula the pullback density of μ_0 under G is $\nu(y) = \mu_0(G(y))|\det J_G(y)|$, so the log-ratio between the target and the model along the flow is the single scalar field

$$z(y) := \log \frac{\mu(y)}{\nu(y)} = U_0(G(y)) - U(y) - \log |\det J_G(y)| + \text{const}, \quad y \in \mathbb{R}^d. \quad (21)$$

Both ingredients of the loss are elementary functionals of z . The forward KL is its mean under the target, $\text{KL}(\mu\|\nu) = \mathbb{E}_{y \sim \mu}[z(y)] + \text{const}$ (cf. (2)), and each cross-regularizer X_{ω} (3) is the mean absolute pairwise variation of z under a non-negative weight ω , $X_{\omega}(\mu\|\nu) = \mathbb{E}_{y, y' \sim \omega}[|z(y) - z(y')|]$. Combining the shape term $\omega = \mu$ with the balanced mixture $\omega = \alpha\hat{\mu} + \beta\bar{\nu}$ that handles mode discovery and leakage suppression, the total training loss in terms of the flow map G is

$$\mathcal{L}[G] = \underbrace{\mathbb{E}_{y \sim \mu}[z(y)]}_{\text{KL}(\mu\|\nu)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu}[|z(y) - z(y')|]}_{X_{\mu}(\mu\|\nu)} + \underbrace{\mathbb{E}_{y, y' \sim \alpha\hat{\mu} + \beta\bar{\nu}}[|z(y) - z(y')|]}_{X_{\alpha\hat{\mu} + \beta\bar{\nu}}(\mu\|\nu)}. \quad (22)$$

We stress the two pushforward measures appearing in (22). The model distribution $\nu = G_{\#}^{-1}\mu_0$, with density $\nu(y) = \mu_0(G(y))|\det J_G(y)|$, is the flow’s live approximation to the target μ and depends on the trainable parameters of G . The barred measure $\bar{\nu}$ is the *same* pushforward frozen at the current parameters – a stop-gradient copy of ν – which enters the mixture term only as a sampling weight and is never differentiated through.

All three terms are computed in a single forward pass: evaluate $\mathbf{z} = (z(y_1), \dots, z(y_B))$ once on a batch, take $\text{mean}(\mathbf{z})$ for the forward KL, and $\text{mean}(|\mathbf{z} - \mathbf{z}[\sigma]|)$ for a random permutation σ of the batch for each X term. Backpropagation passes through $\mathbf{z}$ exactly once, identical to a vanilla forward KL step, so the X functionals add essentially no extra cost.

The loss draws on three measures, of which only two must be built. The $\bar{\nu}$ batch is free: it is the raw pushforward $\bar{y}_i = G^{-1}(x_i)$ of source samples $x_i \sim \mu_0$, taken under `torch.no_grad`; this detached sampler keeps the loss numerically stable on sharp targets, where differentiating through the sampling chain (as a true reverse-KL gradient does) would explode. The other two require construction: samples of the target μ for the forward KL and X_{μ} terms, supplied by annealed importance sampling (Section 3.2), and samples of the wide-coverage measure $\hat{\mu}$ for the mixture term, supplied by quench and temper (Section 3.3).

Although (λ, α, β) are three free parameters in principle, the X functional is by construction on the same scale as the forward KL, so any choice of order 1 works in practice. Two regimes are worth highlighting. The simple regime sets $\lambda = 1$ and $\alpha = \beta = 0$, adding only X_{μ} on top of the forward KL; this term reuses the AIS samples and autograd graph of the forward KL, costing only an extra permutation and elementwise difference per batch, and it suffices whenever the source μ_0 already overlaps every mode of μ . The mode-discovery regime sets $\lambda = 1$ and $\alpha = \beta = 1/2$, turning on the mixture term with a balanced half-half draw from $\hat{\mu}$ and $\bar{\nu}$: the $\hat{\mu}$ half pulls ν toward QT-discovered modes that AIS alone cannot reach, while the $\bar{\nu}$ half restrains the mass the flow places outside the support of μ . We recommend the mode-discovery regime as the default whenever μ_0 may not cover all modes of μ , and use it throughout the experiments; re-tuning (λ, α, β) away from these defaults did not improve results on any of our benchmarks.

3.2 Sampling μ by annealed importance sampling

Evaluating the μ -expectations in (22) requires samples from μ , which are not directly available. The natural surrogate draws $y_1, \dots, y_B$ from the pushforward ν and reweights by

$$w(y) = \frac{\mu(y)}{\nu(y)} = \exp(z(y)) = \exp\left(U_0(G(y)) - U(y) - \log |\det J_G(y)|\right), \quad y \in \mathbb{R}^d, \quad (23)$$

followed by a resampling step and a short Langevin simulation to diffuse the duplicates. In high dimensions, however, μ and ν typically differ greatly, so the weights $w(y_i)$ collapse onto a few outliers and the resulting samples are of very low quality. Annealed importance sampling (AIS) bridges this gap by inserting $M + 1$ intermediate distributions

$$\pi_k(y) = \nu^{1-\frac{k}{M}} \mu^{\frac{k}{M}} = \nu(y) \cdot w(y)^{\frac{k}{M}}, \quad k = 0, 1, \dots, M, \quad (24)$$

so that $\pi_0 = \nu$ and $\pi_M = \mu$. Each rung of the ladder reweights by only $w(y)^{\frac{1}{M}}$, keeping the weight ratio close to 1 and avoiding the degeneracy of direct importance sampling:

Algorithm 1: Annealed importance sampling from ν to μ

Input: samples $y_1, \dots, y_B$ from ν , ladder length M

Output: samples $y_1, \dots, y_B$ approximately distributed as μ

```

1 for  $k = 1, \dots, M$  do
2   Compute the per-step weights  $w(y_i)^{\frac{1}{M}}$  on  $y_1, \dots, y_B$ ;
3   Resample  $y'_1, \dots, y'_B$  from the weighted empirical distribution;
   // The resampled set has duplicates; Langevin diffuses them
4   Apply a short Langevin simulation to  $y'_1, \dots, y'_B$ , returning fresh samples  $y_1, \dots, y_B \sim \pi_k$ ;
5 end
```

The Langevin step nominally targets the intermediate distribution π_k , and so requires its score $\nabla_y \log \pi_k(y)$. Through the pushforward density $\nu(y) = \mu_0(G(y)) |\det J_G(y)|$, this score carries the term $\nabla_y \log |\det J_G(y)|$, whose evaluation needs the Jacobian $J_G(y)$ itself – not merely its determinant, which the flow already provides – and is therefore expensive for a deep map. In practice we instead diffuse the resampled particles directly under the target μ , whose score $\nabla_y \log \mu(y) = -\nabla U(y)$ is cheap. Since the per-rung reweighting already follows the geometric path (24) and μ is the ultimate target, this substitution introduces no essential bias: the Langevin kernel serves only to decorrelate the duplicates left by resampling.

Remark (Choosing the ladder length M) For simple targets close to the source, $M = 1$ already suffices; in high dimensions or under large deformations, $M = d$ is a good default. For instance, annealing $\mu_0 \propto e^{-|x|^2}$ to $\mu \propto e^{-2|x|^2}$ on $\mathbb{R}^d$, a ladder of $M = d$ steps keeps the weighted empirical distribution at a finite ESS that does not degrade with d .

3.3 Constructing $\hat{\mu}$ by quench and temper

The mixture term needs samples from the wide-coverage measure $\hat{\mu}$. These are prepared once, before training, as a fixed pool of P points $x_1, \dots, x_P$ distributed as $\hat{\mu}$; at each training step a mini-batch is drawn from the pool and refreshed by a short Langevin chain before entering the X_ω computation of Section 3.1 with μ replaced by $\hat{\mu}$. The pool is produced by quench and temper (QT), which proceeds in three stages: diffusion (melt), optimization (quench), and rejuvenation (temper).

Algorithm 2: QT: initiative mode discovery

Input: samples $x_1, \dots, x_P$ from μ_0
Output: samples $x_1, \dots, x_P$ forming $\hat{\mu}$
 // Diffusion (melt): scatter the samples to a wider region of $\mathbb{R}^d$
 1 Choose a large $\sigma > 0$ and update $x_i \leftarrow x_i + \sigma \xi_i$ with $\xi_i \sim \mathcal{N}(0, I_d)$ i.i.d.;
 // Optimization (quench): drive each sample toward a mode center of μ
 2 Run LBFGS on U for a fixed number of steps to update $x_1, \dots, x_P$;
 // Rejuvenation (temper): spread the samples around each mode
 3 Run a short Langevin simulation targeting μ on $x_1, \dots, x_P$;

The output samples concentrate near the mode centers of μ , and the initial Brownian diffusion ensures that even modes far from the support of the input are reachable. The $X_{\hat{\mu}}$ regularizer therefore penalizes any mode of μ on which the current pushforward ν places too little mass, driving the flow to discover modes that AIS alone could never reach.

A large diffusion σ biases the QT output toward modes that lie far from the bulk of μ_0 . This is desirable rather than problematic: the modes the current pushforward fails to cover are precisely those far from μ_0 , and the bias is also harmless because the minimizer of $X_{\hat{\mu}}$ is insensitive to the precise shape of $\hat{\mu}$.

Remark (QT failure mode: heavy-tailed manifolds) *On heavy-tailed manifold targets the thin outer tail consists of narrow-variance components whose basins are too small for the LBFGS quench to land in: the diffusion step rarely scatters particles close enough, and any near miss is pulled back into the wide-basin body. The resulting $\hat{\mu}$ traces the body but misses the tail, and $X_{\hat{\mu}}$ inherits the blind spot.*

3.4 Importance sampling at evaluation: ESS and coverage

At evaluation time the trained flow G supplies the model distribution $\nu = G_{\#}^{-1} \mu_0$ as an approximation to μ , together with the exact tractable density $\nu(y) = \mu_0(G(y)) |\det J_G(y)|$ that any normalizing flow makes available. Because the density is exact, the importance weight $w(y) = \mu(y)/\nu(y) = \exp(z(y))$ of (23) can be evaluated for free on any pushforward sample and used at test time to correct the residual bias of ν as an approximation to μ . Drawing $\mathbf{y} = (y_1, \dots, y_N)$ with $y_i = G^{-1}(x_i)$ and $x_i \sim \mu_0$ i.i.d., the self-normalized importance-sampling estimator

$$\widehat{\mathbb{E}_{\mu}[f]} = \frac{\sum_{i=1}^N w(y_i) f(y_i)}{\sum_{i=1}^N w(y_i)} \quad (25)$$

is a consistent estimator of $\mathbb{E}_{\mu}[f]$ for any μ -integrable test function f , regardless of how far ν is from μ : the flow does not need to be exact for the estimate to converge as $N \rightarrow \infty$. To obtain approximate samples from μ rather than just expectations one resamples $\mathbf{y}$ by the weights and refreshes with a short Langevin chain targeting μ , exactly the surrogate that drove the AIS step in Section 3.2 but now applied with the trained flow held fixed. The training bias the X functionals were designed to reduce is, at this stage, traded for sampling cost: the better the flow, the smaller the residual reweighting that (25) has to do, and the lower its variance.

Remark (Coarse training, fine importance sampling) *If the target has a natural coarse representation, the flow can be trained on the cheap low-dimensional coarse version and the resulting proposal reweighted via (25) against the fine high-dimensional target at evaluation – the high-dimensional target then never enters a gradient step. This separation is the way Lin et al. (2026) applies the framework to path integral Monte Carlo.*

We assess the quality of this pipeline through two complementary metrics: the effective sample size (ESS), which measures how uniform the importance weights are on the support reached by the flow, and the coverage, which checks whether every mode of μ has been reached at all.

The first metric is the effective sample size. Let $\mathbf{y} = (y_1, \dots, y_N) \in \mathbb{R}^{N \times d}$ be i.i.d. samples from ν , with importance weights $w(y_i)$ given by (23). The ESS is defined as

$$\text{ESS}(\mathbf{y}) = \frac{\left(\sum_{i=1}^N w(y_i)\right)^2}{N \sum_{i=1}^N w(y_i)^2} \in [0, 1], \quad (26)$$

where the upper bound 1 follows from the Cauchy–Schwarz inequality and is attained when all weights are equal. When G is perfectly trained the weights are constant and the ESS reaches 1. As emphasized in the introduction, however, this conclusion is only one-way: a high ESS certifies that the weights are uniform *on the support of ν* , but modes of μ that the pushforward never reaches contribute nothing to the sum and so cannot lower the ESS, which gives rise to the *fake ESS* pitfall.

To detect such missing modes, we pair the ESS with the coverage metric of Naeem et al. (2020). Let $\mathbf{x} = (x_1, \dots, x_P) \in \mathbb{R}^{P \times d}$ be the samples of $\hat{\mu}$ produced by the quench and temper algorithm (Algorithm 2). For each $1 \leq i \leq P$, the k -nearest-neighbor radius of x_i is

$$\text{NND}_k(x_i; \mathbf{x}) = \inf \left\{ r \geq 0 : \#\{1 \leq j \leq P : 0 < |x_i - x_j| < r\} \geq k \right\}, \quad (27)$$

i.e., the distance from x_i to its k -th nearest neighbor in $\mathbf{x}$. The coverage of $\mathbf{x}$ by the pushforward samples $\mathbf{y}$ is then

$$\text{Coverage}_k(\mathbf{y}; \mathbf{x}) = \frac{1}{P} \sum_{i=1}^P \mathbf{1} \left\{ \exists 1 \leq j \leq N : |y_j - x_i| \leq \text{NND}_k(x_i; \mathbf{x}) \right\}, \quad (28)$$

i.e., the fraction of points in $\mathbf{x}$ whose k -nearest-neighbor ball contains at least one pushforward sample. A coverage close to 1 certifies that every region populated by $\hat{\mu}$ has been reached by the flow, while a coverage substantially below 1 exposes precisely the missing modes that the ESS alone cannot see.

4 Numerical experiments

4.1 2D benchmark distributions

We test the training scheme on six 2D targets, each chosen to isolate a different failure mode of the forward KL rather than to match the GMM/DW community suite (Akhound-Sadegh et al., 2024; Woo and Ahn, 2024): Threewell (three asymmetric wells of a quartic-plus-sinusoid potential, with the narrow source far from all three), Himmelblau (four isolated modes), Annulus (a thin one-dimensional ring of minima together with an isolated central minimum), Python (a thin 1D-manifold version of the Python logo built from 1024 stroke Gaussians traced along the skeleton of the two interlocking snake bodies, plus two larger eye Gaussians), Chessboard (a Gaussian mixture on 26 of the 32 black squares of an 8×8 grid, with six cells removed to break the chessboard symmetry), and Sparse (a four-mode Gaussian mixture with a dense pair near the origin and two isolated modes in opposite corners, several source-standard-deviations away, that the narrow source tail cannot reach). The potentials are listed in Table 1.

name	potential $U(x_1, x_2)$	minima
Threewell	$6[(x_1^2 - 1)^2 + (x_2^2 - 1)^2 + \sin(x_1 + 2x_2)]$	three asymmetric wells
Himmelblau	$(x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$	four isolated points
Annulus	$20(r^6 - 8r^4 + 16r^2 + 1)^{1/3}$, $r^2 = x_1^2 + x_2^2$	$r = 0$ and ring at $r = 2$
Python	Python-like Gaussian mixture	1024 stroke centers + 2 eye centers
Chessboard	Chessboard-like Gaussian mixture	26 centers
Sparse	Sparse Gaussian mixture	2 dense + 2 isolated centers

Table 1: Six 2D benchmark targets.

For all six targets the source μ_0 is an isotropic Gaussian $\mathcal{N}(0, \sigma^2 I)$ with σ between 0.25 and 2.0; on Threewell it is deliberately narrow and centered at the origin, far from all three wells, so the initial pushforward covers none of them. The flow is a neural spline flow with 6 coupling transforms, 32 bins, and hidden widths (128, 128), on a square box matched to each target. Training uses Adam (learning rate 10^{-3} to 5×10^{-3}) for 1000 steps (2000 on Python, whose narrow tubular support takes longer to saturate), with batch size 500, enlarged to 1000–2000 on the multi-mode Threewell, Chessboard, and Sparse so a single batch can cover all modes. At each step the AIS surrogate is one importance-weighted resampling pass against the current pushforward followed by 50 Langevin steps. The loss coefficients are the preferred default of Section 3.1, $\lambda = 1$ and $\alpha = \beta = 1/2$, and the wide-coverage measure $\hat{\mu}$ is built once before training by QT (Algorithm 2) and recycled with one Langevin rejuvenation per step.

For each target we compare four losses obtained by progressively adding one X regularizer at a time: the bare forward KL (2); $\text{KL} + X_\mu$ with $\lambda = 1$; $\text{KL} + X_\mu + X_{\hat{\mu}}$ with $\alpha = 1$, $\beta = 0$; and the preferred mixture $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ with $\alpha = \beta = 1/2$.

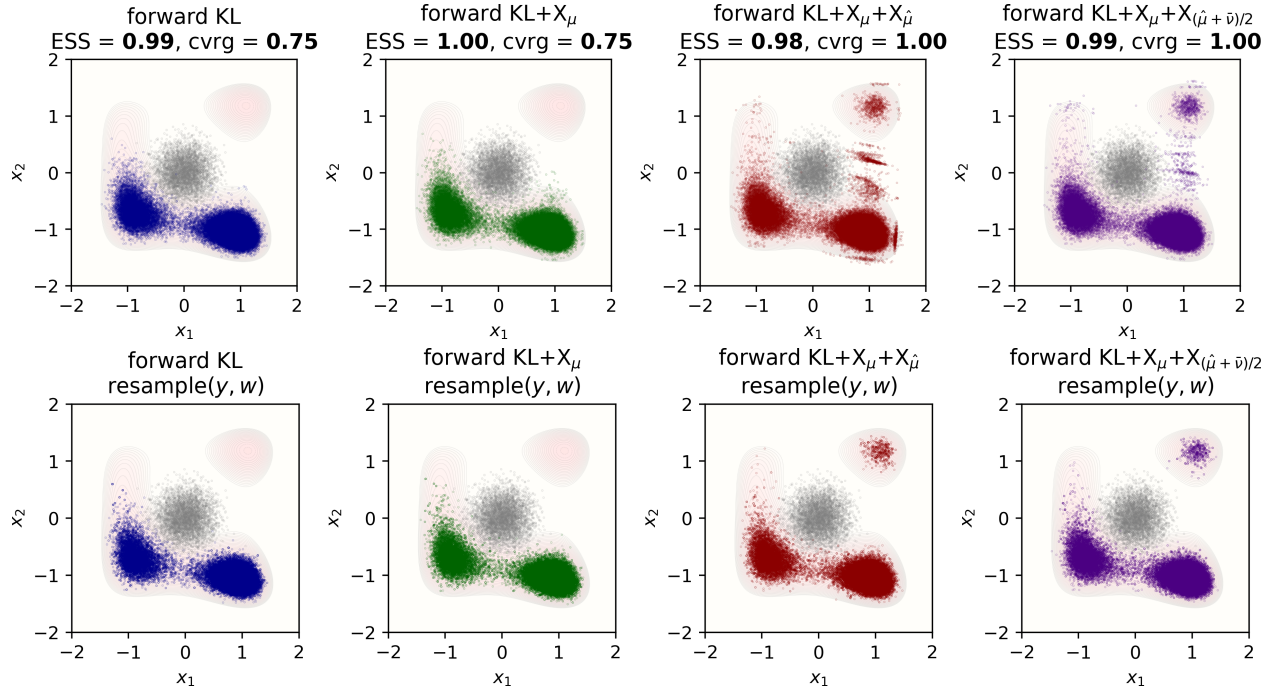
Threewell. The quartic-plus-sinusoid potential has three asymmetric wells, and the narrow source μ_0 (standard deviation 0.25) sits at the origin, far from all of them, so the initial pushforward covers none. Forward KL and $\text{KL} + X_\mu$ lock onto only two of the three wells (coverage ≈ 0.75 in both cases) and never discover the third, yet their ESS remains close to one (0.99 and 1.00): a fake-ESS instance in which the flow omits an entire mode while reporting near-uniform weights on the support it does reach. Adding $X_{\hat{\mu}}$ pulls mass onto the QT-discovered third well and lifts coverage to ≈ 1.00 , and the mixture variant $X_{(\hat{\mu} + \bar{\nu})/2}$ attains the same full coverage at the highest full-coverage ESS (0.99) with the cleanest wells of the four (Figure 1a). On Threewell, $X_{\hat{\mu}}$ and the mixture extension are the only two losses that recover the missed third well.

Himmelblau. The source places nearly all its mass near the central saddle, so the initial pushforward is far from any of the four Himmelblau modes. Forward KL and $\text{KL} + X_\mu$ both converge to a flow that covers only two of the four modes (coverage ≈ 0.53 and ≈ 0.48), yet the ESS gives no warning: $\text{KL} + X_\mu$ posts the highest nominal ESS of the four (0.97) while missing half the modes, a fake-ESS instance. Adding $X_{\hat{\mu}}$ steers mass toward the QT-discovered modes and recovers coverage 1.00 (Figure 1b); the mixture variant $X_{(\hat{\mu} + \bar{\nu})/2}$ also reaches coverage 1.00, and the two full-coverage losses tie at the best full-coverage ESS (0.95), with the mixture pushforward the visibly cleaner of the two. This case confirms the role of $X_{\hat{\mu}}$ (or its mixture extension) as the safeguard one needs precisely when μ_0 does not by itself cover all modes of μ , the regime in which the forward KL is silent.

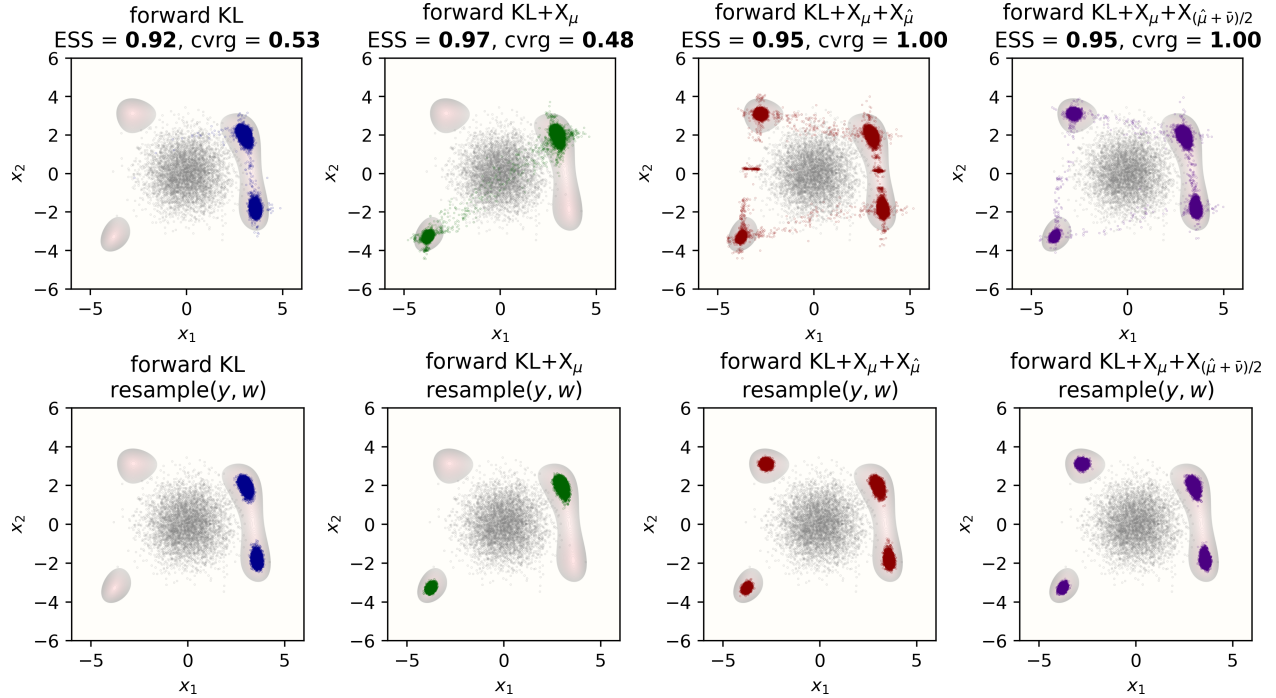
Annulus. All four methods reach coverage 1.00, so $X_{\hat{\mu}}$ contributes no mode rescue here and the discriminating quantities are the flow shape and ESS. Forward KL alone settles at $\text{ESS} \approx 0.66$, leaving visible axis-aligned filaments across the central barrier. Adding X_μ tightens the pushforward to a clean ring-plus-point geometry and lifts the ESS to ≈ 0.97 ; appending $X_{\hat{\mu}}$ on top slightly degrades it to ≈ 0.94 , while the mixture $X_{(\hat{\mu} + \bar{\nu})/2}$ attains the best ESS of the four (≈ 0.97 , Figure 2a). This case isolates the role of X_μ as a shape regularizer: it pointwise penalizes log-ratio variations that the global forward KL leaves untouched.

Python. The Python target lives on an essentially one-dimensional manifold, the skeleton of the two interlocking snake bodies of the Python logo, traced by 1024 stroke Gaussians of width $\sigma_{\text{stroke}} = 0.04$ plus two larger eye Gaussians of width $\sigma_{\text{eye}} = 0.30$. All four methods reach coverage 1.00, so $X_{\hat{\mu}}$ contributes no mode rescue. The ESS plateaus well below 1: bare forward KL settles at ≈ 0.62 , adding X_μ lifts it to ≈ 0.75 , appending $X_{\hat{\mu}}$ holds it at ≈ 0.73 , and the mixture $X_{(\hat{\mu} + \bar{\nu})/2}$ reaches the best ESS of the four, ≈ 0.84 (Figure 2b). On this manifold-dominated target the mode-discovery role of $X_{\hat{\mu}}$ is inactive, yet the X-augmented losses still improve on bare forward KL; the mixture leads the next-best X variant by ≈ 0.10 in ESS, consistent with $\bar{\nu}$ anchoring the off-manifold profile of ν .

Chessboard. Coverage is 1.00 for all four methods, but the visible mode structure and ESS vary. Forward KL alone scatters mass loosely across the black squares, with diffuse bridges between cells and leakage into the six removed cells ($\text{ESS} \approx 0.77$); adding X_μ sharpens the in-cell clusters but barely moves the ESS

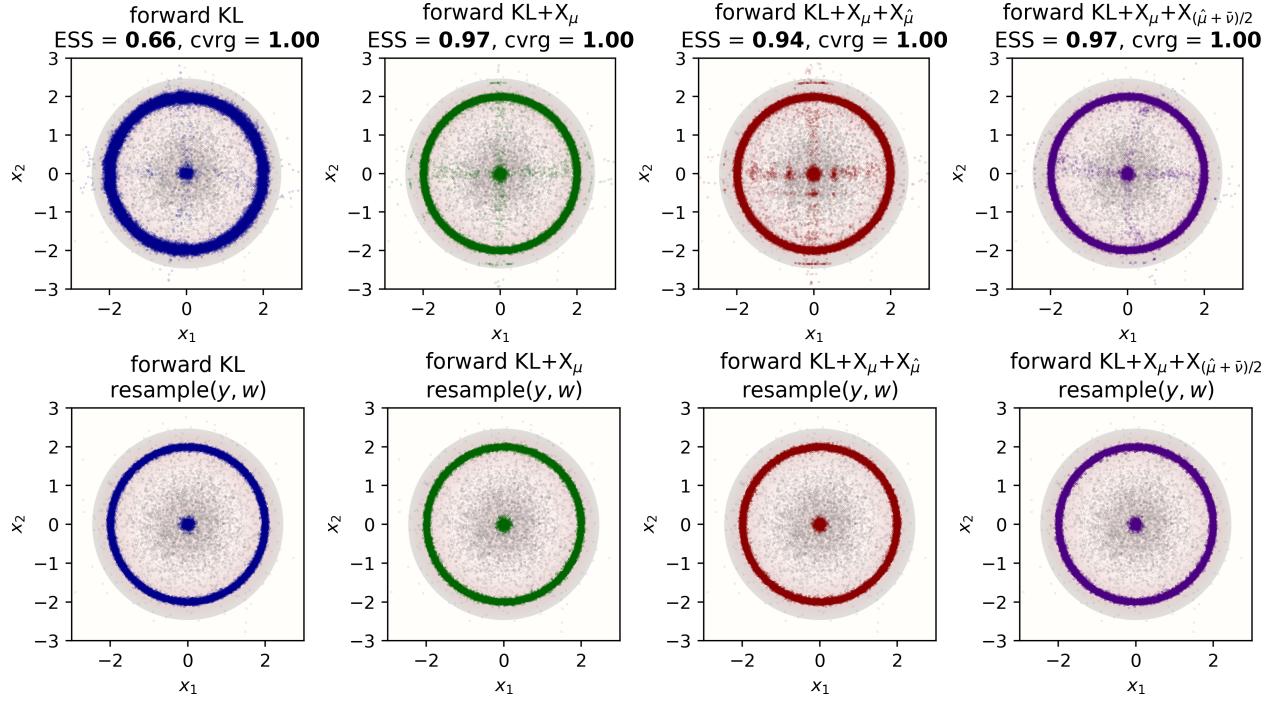


(a) Threewell target.

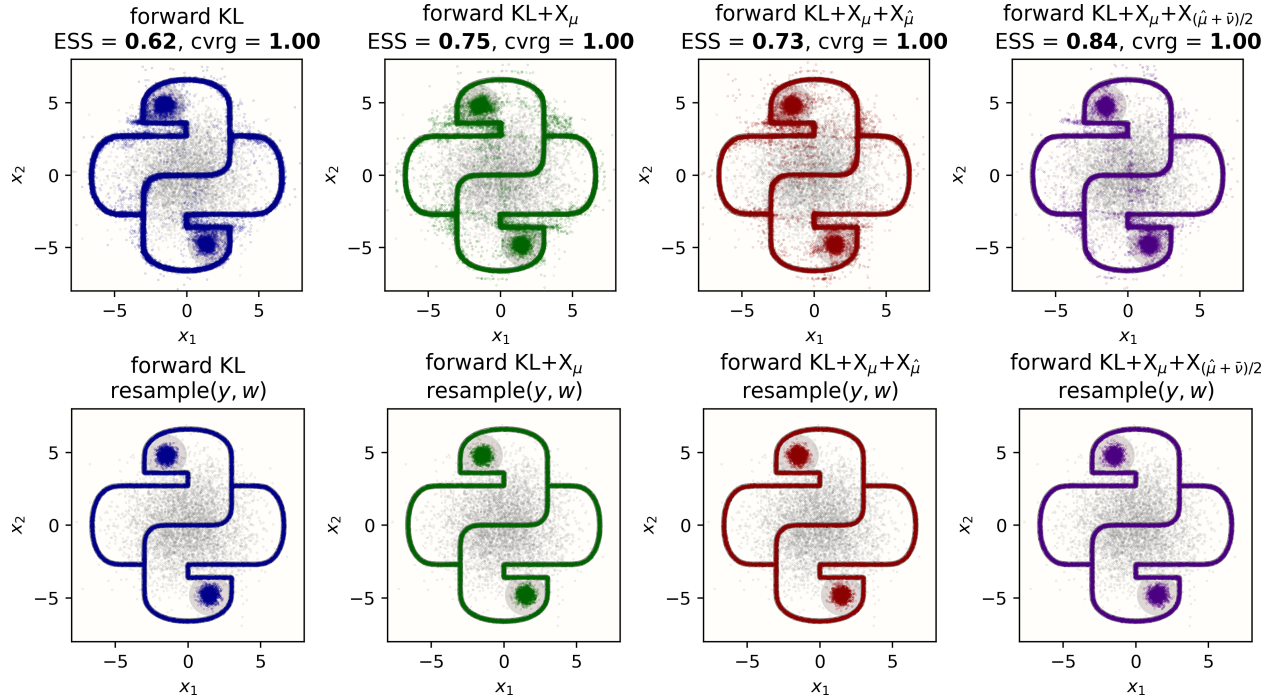


(b) Himmelblau target.

Figure 1: Mode-discovery targets. In each panel, Row 1 is the pushforward $y = G^{-1}(x)$ and Row 2 is $\text{resample}(y, w)$ with Langevin rejuvenation. Columns left to right: forward KL, KL+ X_{μ} , KL+ $X_{\mu}+X_{\hat{\mu}}$, KL+ $X_{\mu}+X_{(\hat{\mu}+\bar{v})/2}$. Each pushforward title reports the final ESS and the $k = 5$ coverage against $\hat{\mu}$. Forward KL and KL+ X_{μ} miss modes at near-perfect ESS, while $X_{\hat{\mu}}$ and the mixture recover full coverage.



(a) Annulus target.



(b) Python target.

Figure 2: Shape and manifold targets. Rows and columns as in Figure 1.

(≈ 0.76) and does not stop the leakage, and adding $X_{\hat{\mu}}$ on top lifts it only marginally (≈ 0.78), since $\hat{\mu}$ alone is sample-blind to the removed cells. The mixture variant $X_{(\hat{\mu}+\bar{\nu})/2}$ resolves this: the $\bar{\nu}$ half of the sampling sees the holes whenever ν leaks mass there, and the resulting cross contributions $|\log(\mu/\nu)(y) - \log(\mu/\nu)(y')|$ with $y \sim \hat{\mu}$ and $y' \sim \bar{\nu}$ anchor in-cell and out-of-cell log-ratios to each other. The trained flow then places mass cleanly on the 26 filled cells and attains the best ESS of the four by a clear margin (≈ 0.84 , Figure 3).

Sparse. A four-mode Gaussian mixture with a dense pair at the origin and two isolated modes in opposite corners; the source μ_0 (standard deviation 0.6) is narrow enough that its tail never reaches the corners, so the initial pushforward covers only the central pair. Forward KL and $KL+X_{\mu}$ lock onto the two central modes and never discover the isolated corners (coverage ≈ 0.50 in both cases), yet their ESS stays high and stable (0.88 and 0.94): another fake-ESS instance, in which collapsing onto the easy half of the support is rewarded with near-uniform weights. Adding $X_{\hat{\mu}}$ pulls mass onto the QT-discovered corner modes and restores coverage 1.00, but at a cost: lacking the $\bar{\nu}$ half, it overfits the wide-coverage measure $\hat{\mu}$ and leaves visible artificial bridges of mass between the modes (ESS ≈ 0.87). The mixture variant $X_{(\hat{\mu}+\bar{\nu})/2}$ also reaches coverage 1.00 while removing these bridges, placing clean mass on all four modes at the best full-coverage ESS (≈ 0.93 , Figure 4).

target	KL	KL+ X_{μ}	KL+ X_{μ} + $X_{\hat{\mu}}$	KL+ X_{μ} + $X_{(\hat{\mu}+\bar{\nu})/2}$
Threewell	0.99/0.75	1.00/0.75	0.98/ 1.00	0.99/1.00
Himmelblau	0.92/0.53	0.97/0.48	0.95/1.00	0.95/1.00
Annulus	0.66/1.00	0.97/1.00	0.94/1.00	0.97/1.00
Python	0.62/1.00	0.75/1.00	0.73/1.00	0.84/1.00
Chessboard	0.77/1.00	0.76/1.00	0.78/1.00	0.84/1.00
Sparse	0.88/0.50	0.94/0.50	0.87/ 1.00	0.93/1.00

Table 2: Final ESS / $k = 5$ coverage against $\hat{\mu}$ for each loss on each 2D target (best full-coverage ESS per row in bold). On the mode-discovery targets (Threewell, Himmelblau, Sparse) forward KL and X_{μ} post high ESS while silently missing modes, and only $X_{\hat{\mu}}$ and the mixture restore full coverage. On the remaining targets coverage is complete and X_{μ} improves the ESS except on Chessboard, where only the mixture does. The mixture is the only variant that attains the best (or tied-best) full-coverage ESS on every target while always retaining full coverage.

Across the six targets two regimes emerge. On the mode-discovery targets (Threewell, Himmelblau, Sparse) the ESS stays high for every loss and is therefore useless on its own: forward KL and $KL+X_{\mu}$ post ESS up to 1.00 while silently missing whole modes, and only the $X_{\hat{\mu}}$ and mixture variants restore full coverage. On the shape, manifold, and leakage targets (Annulus, Python, Chessboard) coverage is already complete and ESS becomes the discriminator: X_{μ} helps on Annulus and Python but is flat on Chessboard, whereas the mixture improves on bare forward KL by +0.31, +0.22, and +0.07 and attains the best ESS on all three. Overall the mixture is the only loss among the four compared that reaches full mode coverage and the best (or tied-best) full-coverage ESS on every target, and its ESS curves are visibly smoother than bare forward KL.

The mixture loss $\mathcal{L} = KL(\mu\|\nu) + X_{\mu}(\mu\|\nu) + X_{(\hat{\mu}+\bar{\nu})/2}(\mu\|\nu)$ with $\lambda = 1$ and $\alpha = \beta = 1/2$ is the recommended default. It absorbs the mode-discovery role of $X_{\hat{\mu}}$ while suppressing the off-support leakage that $X_{\hat{\mu}}$ alone can introduce when the target has holes interleaved with its modes.

4.2 Bayesian inverse problem: sensor-array source localization

Our next benchmark is a physical Bayesian inverse problem where the multimodality comes from a symmetry of the measurement, not from a hand-built mixture. The setting is source localization, a standard problem in acoustics and signal processing: a line of fixed sensors records the combined signal of several identical point sources, and we infer the source positions from the noisy readings. The unknown is the vector of source positions $\theta = (\theta_1, \dots, \theta_n) \in \mathbb{R}^n$ for $n = 3$ sources. A source at θ_i leaves a Gaussian bump of width $\ell = 0.8$ at each sensor and the sensors read the sum, so with $J = 25$ sensors at fixed positions $s_1, \dots, s_J$ equispaced

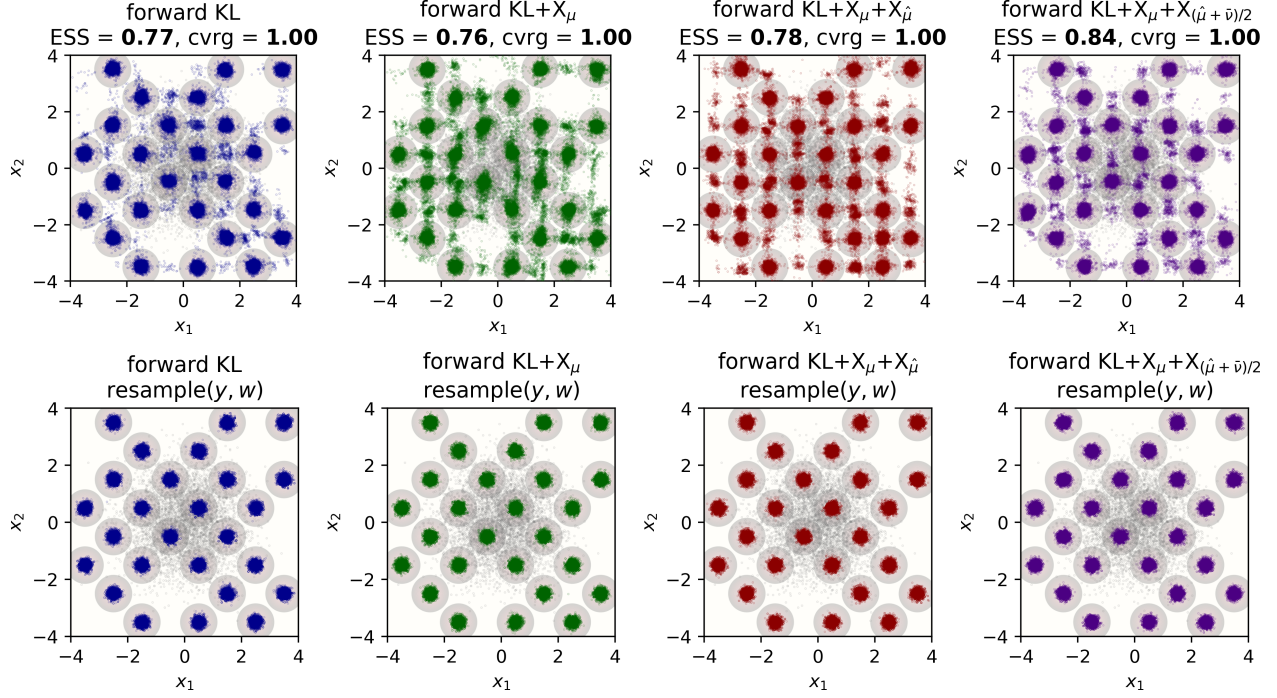


Figure 3: Chessboard target. Row 1: pushforward $y = G^{-1}(x)$. Row 2: $\text{resample}(y, w)$ with Langevin rejuvenation. Columns as in Figure 1.

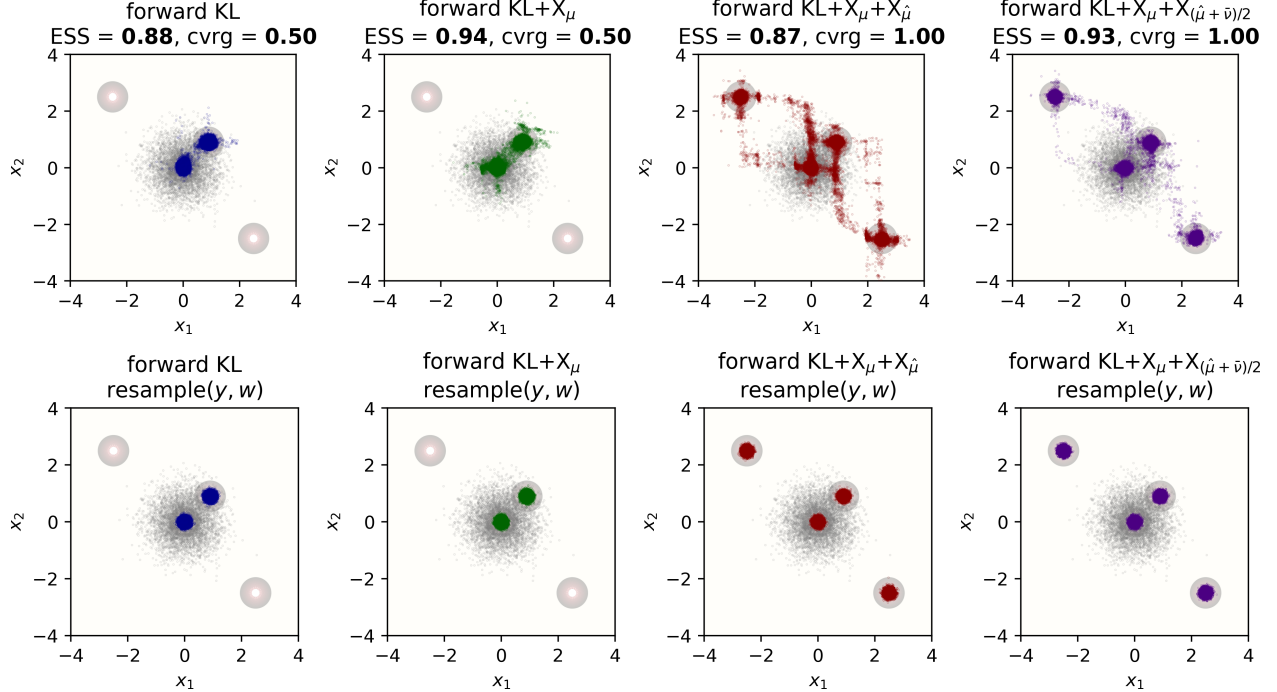


Figure 4: Sparse target. Row 1: pushforward $y = G^{-1}(x)$. Row 2: $\text{resample}(y, w)$ with Langevin rejuvenation. Columns as in Figure 1. Forward KL and $\text{KL}+X_\mu$ collapse onto the two central modes and miss both isolated corners at high, stable ESS (a fake-ESS instance); $X_{\hat{\mu}}$ recovers all four modes but leaves artificial bridges of mass between them, while the mixture recovers them cleanly.

on $[-6, 6]$ the reading at sensor j is

$$g_j(\theta) = \sum_{i=1}^n \exp\left(-\frac{(s_j - \theta_i)^2}{2\ell^2}\right).$$

The data are these readings at the true positions $\theta^* = (5, 0, -5)$ with i.i.d. Gaussian noise,

$$y_j = g_j(\theta^*) + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, \sigma_{\text{obs}}^2), \quad \sigma_{\text{obs}} = 0.3,$$

fixed across all runs. The prior on the positions is a centred isotropic Gaussian $\mu_0 = \mathcal{N}(0, \sigma_{\text{prior}}^2 I_n)$ with $\sigma_{\text{prior}} = 1.6$, which is also the flow’s source, and the negative log-posterior is the prior plus the data fit,

$$U(\theta) = \frac{1}{2\sigma_{\text{obs}}^2} \sum_{j=1}^J (g_j(\theta) - y_j)^2 + \frac{1}{2\sigma_{\text{prior}}^2} \|\theta\|^2. \quad (29)$$

Because the sources are identical, g_j is a sum over them and is unchanged by any permutation of the coordinates, $g_j(\theta_{\pi(1)}, \dots, \theta_{\pi(n)}) = g_j(\theta)$ for $\pi \in S_3$; the prior is permutation-invariant too, so $U(\pi \cdot \theta) = U(\theta)$. The posterior therefore has exactly $3! = 6$ equal, well-separated modes – the permutations of θ^* – and the centred prior sits on their common symmetry point at the origin, about 4.4 prior standard deviations from every mode ($\|\theta^*\| = 7.07$). The prior tail reaches none of them, so a flow trained by bare forward KL only sees the one or two modes its pushforward first lands on and collapses there. We use the same flow, training, and diagnostics as Section 4.1.

Table 3 and Figure 5 report the comparison. Bare forward KL covers only 2 of the 6 modes at ESS 0.90, and adding X_μ posts the highest ESS of the four, 0.98, while collapsing onto a single mode – the ESS improves exactly as coverage gets worse. The QT-driven $X_{\hat{\mu}}$ and the mixture both recover all 6 modes, with mass split evenly, at ESS 0.91 and 0.93.

loss	fresh-pool ESS	modes discovered
forward KL	0.904	2
KL+ X_μ	0.979	1
KL+ X_μ + $X_{\hat{\mu}}$	0.913	6
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.934	6

Table 3: Sensor-array source-localization benchmark (29) with $d = 3$ and $3! = 6$ permutation posterior modes. Bare forward KL and KL+ X_μ post fake-high ESS (the highest, 0.98, for X_μ) while covering only 1–2 of the six equal-mass modes; the QT-driven $X_{\hat{\mu}}$ and mixture variants restore full 6-mode coverage. Modes discovered is the number of the six permutation modes that hold at least 5% of the pushforward mass.

This benchmark is intentionally low-dimensional, and the numbers come from a single training seed; which modes a collapsed method happens to keep is set by the random path, not by the loss.

4.3 High-dimensional product multi-well

To test the framework beyond two dimensions we construct a family of product multi-well targets in dimension $d = 2^k$, playing here the role that double-well and Lennard-Jones systems play in the diffusion-sampler literature (Akhound-Sadegh et al., 2024; Woo and Ahn, 2024), but with an analytically known mode structure and a diagonal optimal transport map so that training failures are unambiguously training failures and not capacity failures of the flow,

$$U(x) = \frac{1}{2} \sum_{i=1}^{2^k} x_i^2 + 12 \sum_{i=1}^k \exp(-x_i^2), \quad (30)$$

a double well in each of the first k coordinates (minima at $\pm\sqrt{\ln 24} \approx \pm 1.78$, separated by a barrier of height ≈ 9.9) and a standard Gaussian in the remaining $2^k - k$. The source is the standard Gaussian $\mu_0 = \mathcal{N}(0, I_d)$,

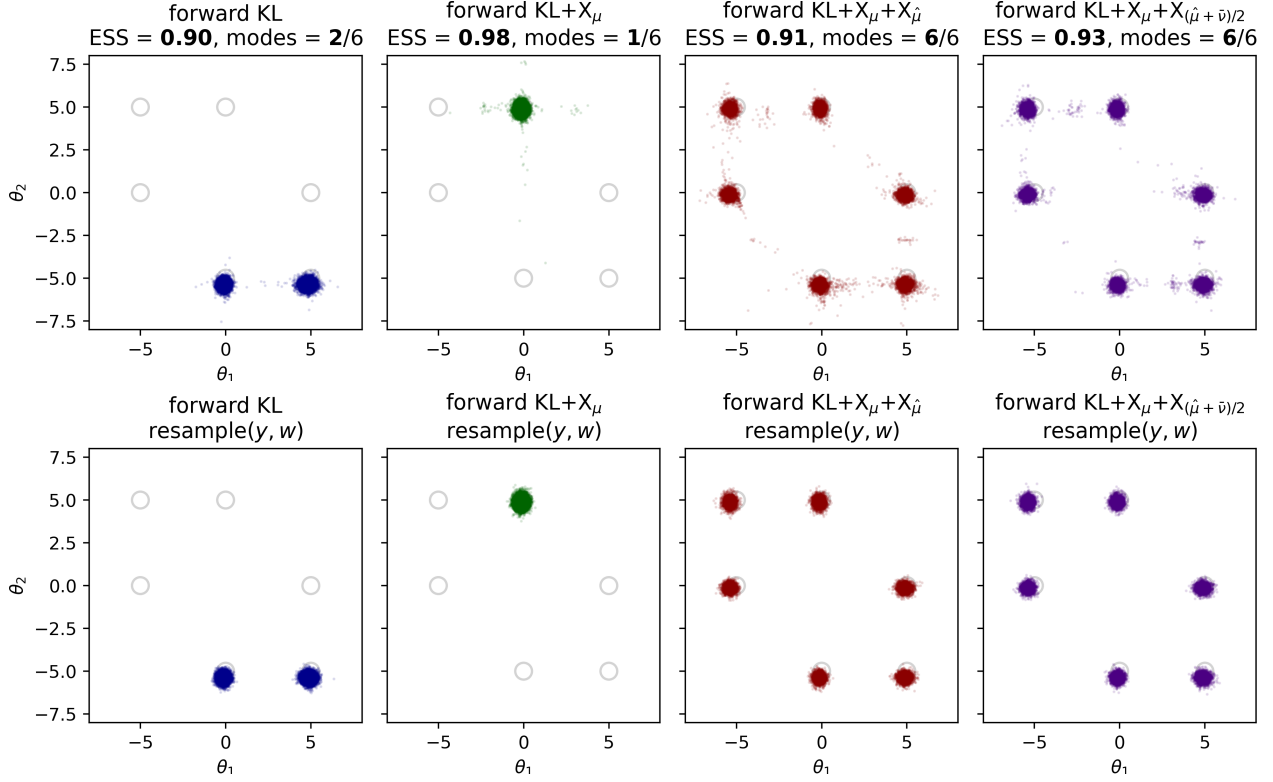


Figure 5: Sensor-array posterior, pushforward positions projected onto (θ_1, θ_2) . Row 1: raw pushforward $y = G^{-1}(x)$; Row 2: $\text{resample}(y, w)$ with Langevin rejuvenation. Columns left to right: forward KL, $\text{KL}+X_\mu$, $\text{KL}+X_\mu+X_{\hat{\mu}}$, $\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$. Light-gray circles mark the six permutation mode centres (the permutations of θ^*); each title reports the final ESS and the modes found out of six. Forward KL and $\text{KL}+X_\mu$ collapse onto 1–2 modes at high ESS, while $X_{\hat{\mu}}$ and the mixture fill all six.

whose potential is the quadratic part of U . Because U is separable, the target factorizes into k identical double wells and $2^k - k$ Gaussians, so it has exactly $2^k = d$ modes – the number of modes equals the dimension – and the optimal transport map is diagonal. The source sits exactly on the barrier of every double well, so the untrained pushforward covers none of the 2^k mode combinations. We sweep $k = 1, \dots, 7$, i.e. $d = 2, 4, 8, 16, 32, 64, 128$.

The flow is a neural spline flow on $[-4, 4]^d$ with 6 coupling transforms, 16 bins, and hidden widths (256, 256). Training uses Adam at learning rate 10^{-3} with batch size 200 – forward KL is highly sensitive to the batch size, so a small batch sharpens the contrast between the losses – drawn from a source pool of $10^4 \cdot 2^k$ samples; the number of steps decreases with d from 3000 ($d \leq 4$) to 1200 ($d \geq 64$). The μ -surrogate is the same one-step importance-resampling pass followed by 50 Langevin steps used in the 2D experiments, and $\hat{\mu}$ is built once by QT from $100 \cdot 2^k$ samples. The four losses are exactly those of Section 4.1, with $\lambda = 1$ and $\alpha = \beta = 1/2$.

Table 4 reports the final ESS of each loss across the eight dimensions. Unlike the 2D mode-discovery targets, here the flow reaches all 2^k modes for every loss – each mode is filled to within a few percent of its uniform 2^{-k} share, and the $k = 5$ coverage against $\hat{\mu}$ stays ≥ 0.94 throughout – so this is the calibration regime of Annulus, Python, and Chessboard rather than the fake-ESS regime, and ESS is the sole discriminator. Three trends are clear. First, the bare forward KL degrades monotonically with dimension, from 0.96 at $d = 2$ to 0.38 at $d = 256$, as the log-weight variance accumulates one double well at a time. Second, the X regularizers counteract this and the gap widens with dimension: negligible at $d = 2$, it reaches +0.21–0.22 at $d = 64$ and 128, and +0.18 at $d = 256$ where the wide-coverage $X_{\hat{\mu}}$ becomes the best loss. Third, the QT-driven losses ($X_{\hat{\mu}}$ and the mixture) overtake X_μ alone at high dimension: X_μ tracks the mixture closely through $d = 64$ but falls back to the bare-KL level by $d = 256$ (0.33 vs 0.38), because it reweights the live

one-step IS surrogate whose pushforward is poorly formed early in training, while $X_{\hat{\mu}}$ and the mixture hold their lead because the fixed QT pool keeps the flow anchored on the wells even as the log-weight variance grows.

Although the nominal dimension is $d = 256$, the *essential* dimension of this target is only 8: the potential (30) is a double well in the first $k = 8$ coordinates and a standard Gaussian in the remaining 248, so all the multimodality lives in an 8-dimensional subspace. Figure 6 sweeps the annealed-importance-sampling ladder length $M \in \{1, 2, 4, 8\}$ that builds the μ -surrogate for the mixture loss at $d = 256$. Increasing M accelerates convergence in the first training stage – a longer ladder leaves the floor sooner – but does not affect the long-time limit: the ESS curves differ only in the first stage and settle at a comparable value. The ladder length is therefore a first-stage accelerator rather than a determinant of the final flow quality.

loss \ d	2	4	8	16	32	64	128	256
forward KL	0.958	0.937	0.935	0.871	0.790	0.669	0.547	0.384
KL+ X_{μ}	0.998	0.983	0.972	0.961	0.893	0.845	0.657	0.333
KL+ X_{μ} + $X_{\hat{\mu}}$	0.987	0.990	0.973	0.955	0.948	0.835	0.750	0.565
KL+ X_{μ} + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.986	0.992	0.975	0.969	0.953	0.878	0.767	0.543

Table 4: Final ESS of each loss on the product multi-well target (30) across dimension $d = 2^k$, $k = 1, \dots, 8$ (best per column in bold). All four losses reach full mode coverage at every d , so ESS is the discriminator. Forward KL degrades steadily with d ; the QT-driven losses ($X_{\hat{\mu}}$ and the mixture) are best or tied-best for every $d \geq 4$. The mixture leads from $d = 4$ to $d = 128$, with a margin of +0.21–0.22 over bare forward KL at $d = 64, 128$; at $d = 256$ the wide-coverage $X_{\hat{\mu}}$ alone takes the lead by +0.18 over bare KL, while X_{μ} alone falls back to the bare-KL level.

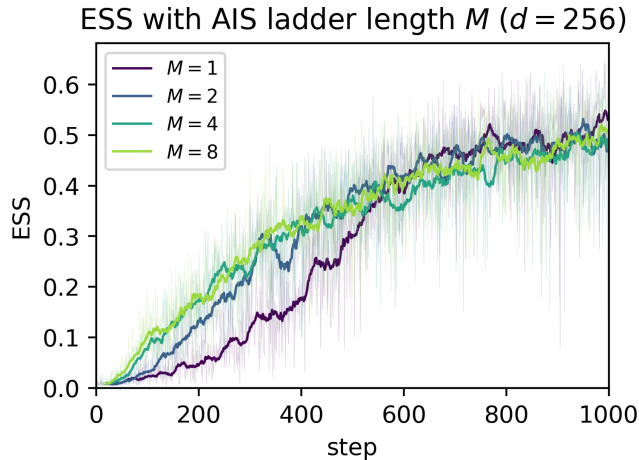


Figure 6: ESS trajectory across training steps for the mixture loss $\text{KL}+X_{\mu}+X_{(\hat{\mu}+\bar{\nu})/2}$ on the product multi-well at $d = 256$, as the annealed-importance-sampling ladder length M that builds the μ -surrogate is varied over $\{1, 2, 4, 8\}$. A longer ladder M accelerates the first-stage convergence – the $M = 1$ one-step surrogate is slowest to leave the floor – but does not change the long-time limit: the curves differ only early and settle at a comparable ESS. Faint lines are the raw per-step ESS; bold lines are a moving average.

References

Tara Akhound-Sadegh, Jarrod Rector-Brooks, Avishek Joey Bose, Sarthak Mittal, Pablo Lemos, Cheng-Hao Liu, Marcin Sendera, Siamak Ravanbakhsh, Gauthier Gidel, Yoshua Bengio, Nikolay Malkin, and Alexander Tong. Iterated denoising energy matching for sampling from Boltzmann densities. In *International Conference on Machine Learning*, 2024.

- Shun-ichi Amari. *Information Geometry and Its Applications*, volume 194 of *Applied Mathematical Sciences*. Springer, 2016.
- Luigi Ambrosio, Nicola Gigli, and Giuseppe Savaré. *Gradient Flows in Metric Spaces and in the Space of Probability Measures*. Lectures in Mathematics ETH Zürich. Birkhäuser, 2 edition, 2008.
- Dominique Bakry, Ivan Gentil, and Michel Ledoux. *Analysis and Geometry of Markov Diffusion Operators*, volume 348 of *Grundlehren der mathematischen Wissenschaften*. Springer, 2014.
- Martin Bauer, Martins Bruveris, and Peter W. Michor. Uniqueness of the Fisher–Rao metric on the space of smooth densities. *Bulletin of the London Mathematical Society*, 48(3):499–506, 2016.
- Arnaud Doucet, Nando De Freitas, and Neil Gordon. An introduction to sequential monte carlo methods. In *Sequential Monte Carlo methods in practice*, pages 3–14. Springer, 2001.
- Marylou Gabrié, Grant M Rotskoff, and Eric Vanden-Eijnden. Adaptive monte carlo augmented with normalizing flows. *Proceedings of the National Academy of Sciences*, 119(10):e2109420119, 2022.
- Richard Jordan, David Kinderlehrer, and Felix Otto. The variational formulation of the Fokker–Planck equation. *SIAM Journal on Mathematical Analysis*, 29(1):1–17, 1998.
- Durk P Kingma and Prafulla Dhariwal. Glow: Generative flow with invertible 1x1 convolutions. *Advances in neural information processing systems*, 31, 2018.
- Guang Lin, Christian Moya, Di Qi, and Xuda Ye. Jeffreys flow: Robust boltzmann generators for rare event sampling via parallel tempering distillation. *arXiv preprint arXiv:2604.05303*, 2026.
- Jun S Liu. *Monte Carlo strategies in scientific computing*, volume 10. Springer, 2001.
- Yulong Lu, Jianfeng Lu, and James Nolen. Accelerating Langevin sampling with birth–death. *arXiv preprint arXiv:1905.09863*, 2019.
- Luca Martino, Víctor Elvira, and Francisco Louzada. Rethinking the effective sample size. *arXiv preprint arXiv:1809.04129*, 2017.
- Laurence Illing Midgley, Vincent Stimper, Gregor NC Simm, Bernhard Schölkopf, and José Miguel Hernández-Lobato. Flow annealed importance sampling bootstrap. *arXiv preprint arXiv:2208.01893*, 2022.
- Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunjey Choi, and Jaejun Yoo. Reliable fidelity and diversity metrics for generative models. In *International conference on machine learning*, pages 7176–7185. PMLR, 2020.
- Radford M. Neal. Annealed importance sampling. *Statistics and Computing*, 11(2):125–139, 2001.
- Frank Noé, Simon Olsson, Jonas Köhler, and Hao Wu. Boltzmann generators: Sampling equilibrium states of many-body systems with deep learning. *Science*, 365(6457):eaaw1147, 2019.
- Felix Otto. The geometry of dissipative evolution equations: the porous medium equation. *Communications in Partial Differential Equations*, 26(1–2):101–174, 2001.
- Felix Otto and Cédric Villani. Generalization of an inequality by Talagrand and links with the logarithmic Sobolev inequality. *Journal of Functional Analysis*, 173(2):361–400, 2000.
- George Papamakarios, Theo Pavlakou, and Iain Murray. Masked autoregressive flow for density estimation. *Advances in neural information processing systems*, 30, 2017.
- Danilo Rezende and Shakir Mohamed. Variational inference with normalizing flows. In *International conference on machine learning*, pages 1530–1538. PMLR, 2015.
- Francisco Vargas, Will Grathwohl, and Arnaud Doucet. Denoising diffusion samplers. In *International Conference on Learning Representations*, 2023.

- Cédric Villani. *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer, 2009.
- Peter Wirnsberger, Andrew J. Ballard, George Papamakarios, Stuart Abercrombie, Sébastien Racanière, Alexander Pritzel, Danilo Jimenez Rezende, and Charles Blundell. Targeted free energy estimation via learned mappings. *The Journal of Chemical Physics*, 153(14), 2020.
- Dongyeop Woo and Sungsoo Ahn. Iterated energy-based flow matching for sampling from Boltzmann densities. In *NeurIPS*, 2024. arXiv:2408.16249.
- Hao Wu, Jonas Köhler, and Frank Noé. Stochastic normalizing flows. *Advances in neural information processing systems*, 33:5933–5944, 2020.
- Qinsheng Zhang and Yongxin Chen. Path integral sampler: A stochastic control approach for sampling. In *International Conference on Learning Representations*, 2022.